

034IN - FONDAMENTI DI AUTOMATICA
- FUNDAMENTALS OF AUTOMATIC CONTROL
ADDITIONAL PARTIAL WRITTEN EXAMINATION
A.Y. 2023/2024

July 19 2024

First and Last Name:

Group: Group A

Exercise: Exercise 1 - 2 Questions

Available Time Frame: 60 minutes

Important Notes:

- Write your answers on a single white sheet of paper using a black pen.
- Do not write in blue ink or pencil.
- Do not hand in additional sheets.
- The clarity and precision of your answers will be subject to evaluation.

I declare that the answers to this exercise are my own and only my own work and that I have not consulted with others.

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Exercise 1

Please refer to the block diagram below:

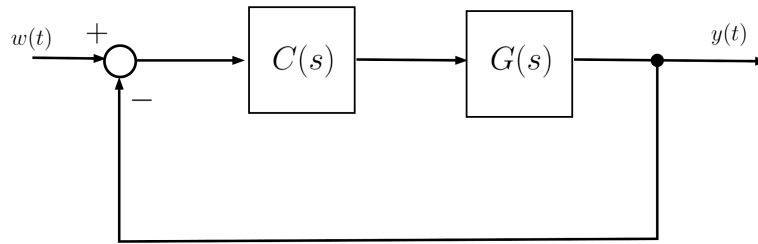


Figura 1: Block diagram representation of an LTI system

where the process is described by the transfer function

$$G(s) = \frac{2}{s^3 + 4s^2 + 5s + 2}$$

and the controller is

$$C(s) = k_P + \frac{k_I}{s}, \quad k_P, k_I \in \mathbb{R}$$

Question 1.1.

- Analyse the **stability** of the closed-loop system as a function of the controller's parameters k_P and k_I .
- Sketch in the plane (k_P, k_I) the **region** corresponding to the **asymptotic stability of the closed-loop system** and the **region** leading to an **unstable closed-loop system**.

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Question 1.2.

- Consider now the scenario when $k_P = 3$, and $k_I = 3$. Analyse in details the stability of the closed-loop system in the present case, applying the Routh-Hurwitz criterion.
- What happens to the Routh table in this scenario? Comment your finding.

Hint: you can also answer this question without having answered the previous question Q1.